

COLLEGE MANAGEMENT SYSTEM

PROJECT REPORT

A Python-based console application for managing students, faculty, courses, attendance, examinations, fees and timetable records.

Project Title	College Management System
Programming Language	Python
Registration Number	26MIM10209
Application Type	Menu-driven Console Application
Core Concepts	Functions, Lists, Loops, Conditional Statements, Input/Output

Submitted as a Python Programming Project

Academic Year: 2026–27

CERTIFICATE

This is to certify that the project titled “**College Management System**”, developed using Python and bearing Registration No. **26MIM10209**, is presented as an academic programming project. The project demonstrates the practical use of Python functions, lists, conditional statements, loops, user input and menu-driven program design.

Project Guide

Student Signature

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my faculty/guide for providing the opportunity to develop this project. This project helped me understand how basic Python programming concepts can be combined to create a practical management application. I also acknowledge the support received during the planning, coding, testing and documentation stages of the project.

ABSTRACT

The **College Management System** is a menu-driven Python application designed to organize common academic records in a simple console environment. The program provides separate modules for student management, faculty management, course management, attendance, examination marks, fees and timetable information. Records are stored in Python lists while functions are used to keep each operation modular and easy to understand.

The application accepts information through the keyboard, performs basic calculations such as attendance percentage, fee balance and examination grades, and allows stored records to be displayed through corresponding view options. The project is intended as an educational demonstration of procedural Python programming and basic data management.

1. OBJECTIVES

- To develop a simple and user-friendly college record management application.
- To practice defining and calling functions in Python.
- To use lists for storing multiple records during program execution.
- To implement menu-driven navigation using loops and conditional statements.
- To calculate attendance percentages and examination grades automatically.
- To calculate fee balance and identify whether a fee record is paid or pending.
- To provide separate options for adding and viewing academic records.

2. SCOPE OF THE PROJECT

The project covers basic academic administration tasks that can be represented using a console-based Python program. It is suitable for demonstrating data entry and retrieval concepts at a small scale. Since records are stored in in-memory lists, the current version is intended primarily for learning and demonstration; records are not permanently saved after the program terminates.

3. TECHNOLOGY AND CONCEPTS USED

Technology / Concept	Purpose in the Project
Python	Primary programming language used to build the application.
Functions	Separate each management operation into reusable blocks.
Lists	Store student, faculty, course and other records.
if / elif / else	Handle menu choices, grading and fee status.
while loop	Keep the main menu running until the Exit option is selected.
input() / print()	Collect user information and display results.
Arithmetic operations	Calculate attendance percentage and fee balance.

4. SYSTEM MODULES

Module	Main Operations
Student Management	Add and view student records: name, registration number, branch and year.
Faculty Management	Add and view faculty records: name, faculty ID and department.
Course Management	Add and view courses using course code, course name and faculty name.
Attendance Management	Store total/attended classes and calculate attendance percentage.
Examination Management	Store subject marks and automatically assign a grade.
Fees Management	Store total fee, paid amount, balance and payment status.
Timetable Management	Store day, time, subject and room number for classes.

5. PROGRAM DESIGN AND WORKING

When the program starts, it repeatedly displays the main menu. The user selects one of seven management modules or chooses Exit. Each management module contains two basic operations: adding a record and viewing stored records. The selected operation calls a dedicated function.

Data Storage

Seven empty lists are initialized at the beginning of the program: **students**, **faculty**, **courses**, **attendance**, **marks**, **fees** and **timetable**. Each new record is represented as a list and appended to the corresponding collection.

Attendance Calculation

Attendance percentage is calculated using the formula: **(classes attended / total classes) × 100**. The result is rounded to two decimal places for display.

Grade Calculation

The examination module uses conditional ranges: 90 and above gives A+, 75–89 gives A, 60–74 gives B, 40–59 gives C, and below 40 gives Fail.

Fee Calculation

The fee module calculates balance using **total fee – amount paid**. A record is displayed as Paid when the balance is zero or negative; otherwise it is marked Pending.

6. ALGORITHM / FLOW OF THE SYSTEM

1. Start the program.
2. Initialize empty lists for all management categories.
3. Display the College Management System main menu.
4. Accept the user's main-menu choice.
5. Open the selected management module.
6. Display Add and View options for that module.
7. If Add is selected, collect input, create a record and append it to the relevant list.
8. If View is selected, check whether records exist and display them.
9. Return to the main menu.
10. Repeat until the user selects Exit.
11. Display the exit message and terminate the program.

7. TESTING AND OBSERVED OUTPUT

Feature Tested	Sample Input / Action	Observed Result
Student	Anushka Tiwari, 209, AI, 1st	Student added successfully.
Faculty	Sasmita Padhy, 299, Scope	Faculty added successfully.
Course	cse1021, programming, Sasmita Padhy	Course added successfully.
Attendance	50 total, 48 attended	Attendance = 96.0%.
Examination	Hiteshi, Maths, 99/100	Grade = A+.
Fees	Total 200000, paid 200000	Fee record added; balance = 0, status Paid.
Timetable	Monday, 9:00am, Cse, room 217	Class added successfully.

8. ADVANTAGES

- Simple menu-driven interface suitable for beginners.
- Modular design makes the program easier to understand and maintain.
- Automates basic calculations such as attendance, grades and fee balance.
- Provides separate storage for different categories of college information.
- Demonstrates several core Python programming concepts in one project.

9. LIMITATIONS

- Records are stored only in memory and are lost when the program closes.
- There is no login or role-based access control.
- The program does not use a database.
- Input validation is limited; incorrect numeric input can cause an error.
- Records are not linked by unique database keys, so consistency depends on user input.

10. FUTURE ENHANCEMENTS

- Add file handling or a database such as SQLite/MySQL for permanent storage.
- Add search, update and delete operations for every module.
- Add input validation and exception handling.
- Introduce student/faculty login with role-based permissions.
- Create a graphical user interface using Tkinter or a web interface.
- Generate reports such as attendance summaries, mark sheets and fee receipts.

11. CONCLUSION

The College Management System successfully demonstrates how Python can be used to build a practical, menu-driven academic management application. The project combines functions, lists, loops, conditional statements, input/output and basic arithmetic into a single working system. The observed outputs confirm the working of the major modules, including student, faculty, course, attendance, examination, fees and timetable management.

Although the current version is intended for educational use and uses temporary in-memory storage, it provides a strong foundation for a more complete college management application using persistent databases, validation and a graphical or web-based interface.

APPENDIX A — PROJECT OUTPUT SCREENSHOTS

The following screenshots document the execution of the College Management System and are placed at the end of the report as requested.

```
=====
COLLEGE MANAGEMENT SYSTEM
Registration No: 26MIM10209
=====
1. Student Management
2. Faculty Management
3. Course Management
4. Attendance Management
5. Examination Management
6. Fees Management
7. Timetable Management
8. Exit

Enter your choice: 1

1. Add Student
2. View Students
Enter option: 1

--- Add Student ---
Enter student name: Anushka Tiwari
Enter registration number: 209
Enter branch: AI
Enter year: 1st
Student added successfully!

=====
COLLEGE MANAGEMENT SYSTEM
Registration No: 26MIM10209
=====
1. Student Management
2. Faculty Management
3. Course Management
4. Attendance Management
5. Examination Management
6. Fees Management
7. Timetable Management
8. Exit

Enter your choice: 2
```

Screenshot 1 — Student Management: adding a student record.

o. Exit

Enter your choice: 6

1. Add Fee
2. View Fees

Enter option: 1

--- Add Fee Record ---

Enter student name: Shriyam

Enter total fee: 200000

Enter amount paid: 200000

Fee record added successfully!

```
=====
                COLLEGE MANAGEMENT SYSTEM
                Registration No: 26MIM10209
=====
```

1. Student Management
2. Faculty Management
3. Course Management
4. Attendance Management
5. Examination Management
6. Fees Management
7. Timetable Management
8. Exit

Enter your choice: 7

1. Add Class
2. View Timetable

Enter option: 1

--- Add Class ---

Enter day: Monday

Enter time: 9:00am

Enter subject: Cse

Enter room number: 217

Class added successfully!

```
=====
```

Screenshot 2 — Faculty Management and Course Management operations.

Fee record added successfully!

```
=====
COLLEGE MANAGEMENT SYSTEM
Registration No: 26MIM10209
=====
```

1. Student Management
2. Faculty Management
3. Course Management
4. Attendance Management
5. Examination Management
6. Fees Management
7. Timetable Management
8. Exit

Enter your choice: 7

1. Add Class
 2. View Timetable
- Enter option: 1

--- Add Class ---

Enter day: Monday

Enter time: 9:00am

Enter subject: Cse

Enter room number: 217

Class added successfully!

```
=====
COLLEGE MANAGEMENT SYSTEM
Registration No: 26MIM10209
=====
```

1. Student Management
2. Faculty Management
3. Course Management
4. Attendance Management
5. Examination Management
6. Fees Management
7. Timetable Management
8. Exit

Enter your choice: 8

Thank you for using College Management System!

Screenshot 3 — Attendance Management: attendance calculation output.

```
Enter your choice: 2

1. Add Faculty
2. View Faculty
Enter option: 1

--- Add Faculty ---
Enter faculty name: Sasmita Padhy
Enter faculty ID: 299
Enter department: Scope
Faculty added successfully!
```

```
=====
                COLLEGE MANAGEMENT SYSTEM
                Registration No: 26MIM10209
=====
```

```
1. Student Management
2. Faculty Management
3. Course Management
4. Attendance Management
5. Examination Management
6. Fees Management
7. Timetable Management
8. Exit
```

```
Enter your choice: 3
```

```
1. Add Course
2. View Courses
Enter option: 1
```

```
--- Add Course ---
Enter course code: cse1021
Enter course name: programming
Enter faculty name: sasmita padhy
Course added successfully!
```

Screenshot 4 — Examination Management: marks and grade output.


```
=====
COLLEGE MANAGEMENT SYSTEM
Registration No: 26MIM10209
=====
1. Student Management
2. Faculty Management
3. Course Management
4. Attendance Management
5. Examination Management
6. Fees Management
7. Timetable Management
8. Exit

Enter your choice: 4

1. Add Attendance
2. View Attendance
Enter option: 1

--- Add Attendance ---
Enter student name: Aryan
Enter total classes: 50
Enter classes attended: 48
Attendance added successfully!
Attendance Percentage: 96.0 %

=====
```

COLLEGE MANAGEMENT SYSTEM

Screenshot 5 — Fees Management and Timetable Management operations.

```

=====
                COLLEGE MANAGEMENT SYSTEM
                Registration No: 26MIM10209
=====

1. Student Management
2. Faculty Management
3. Course Management
4. Attendance Management
5. Examination Management
6. Fees Management
7. Timetable Management
8. Exit

Enter your choice: 5

1. Add Marks
2. View Marks
Enter option: 1

--- Add Examination Marks ---
Enter student name: Hiteshi
Enter subject: Maths
Enter marks out of 100: 99
Marks added successfully!
Grade: A+

=====
                COLLEGE MANAGEMENT SYSTEM
                Registration No: 26MIM10209
=====

1. Student Management
2. Faculty Management
3. Course Management
4. Attendance Management
5. Examination Management
6. Fees Management
7. Timetable Management
8. Exit

```

Screenshot 6 — Timetable entry and program exit output.